

SIH 1505 — INDUSTRIAL EMERGENCY PRE-PLAN

INCIDENT ID: INC-T-03-12K
DOCUMENT VERSION: v1.0 [AUTHORIZED
(PROTOTYPE DEMO)]

GENERATED: 2026-08-15 17:27:12 UTC
PREPARED BY: SIH-1505 Decision Support Engine

FACILITY: PetroChem Complex Alpha - Unit 04
PROTOTYPE FACILITY REF: PCH-ALPHA-04 (Demo Facility — Non-Statutory Evaluation)

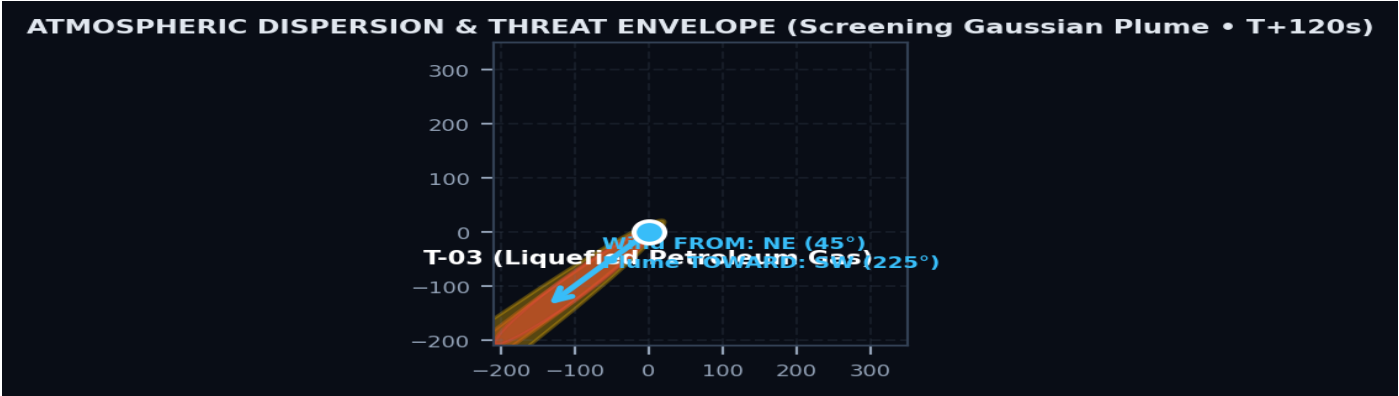
INCIDENT SEVERITY: MODERATE (SCORE: 44.6/100) • GOVERNANCE: AUTHORIZED BY SANTOSH KADAM (Chief Fire Marshal)

1. INCIDENT SOURCE, CHEMICAL SDS & METEOROLOGICAL PROFILE

Incident Type:	Tank Leak	Source Asset:	T-03 (21.6840, 72.5720)
Hazardous Substance:	Liquefied Petroleum Gas (LPG - Propane/Butane)	Release Emission Rate:	25.0 kg/s (Duration: 30 min)
Meteorological Wind Vector:	12.0 km/h • FROM: NE (45°)	Plume Propagation Vector:	Downwind Travel • TOWARD: SW (225°)
Ambient Temperature:	34.0°C	Weather Telemetry Source:	DEMO (Scenario Preset) • Stability: Class D

2. HAZARD THREAT ZONES (SCREENING GAUSSIAN DISPERSION ENVELOPE)

Zone Tier	Threshold Criterion	Downwind Reach	Crosswind Width	Threat Area
Red Threat Zone	ERPG-3 / High Lethality (10000.0 ppm)	318.4 m	24.5 m	5,191 m²
Orange Threat Zone	ERPG-2 / Irreversible Injury (5000.0 ppm)	469.0 m	34.8 m	10,802 m²
Yellow Threat Zone	ERPG-1 / Mild Irritation (1000.0 ppm)	1162.0 m	78.6 m	59,866 m²



3. POPULATION & INFRASTRUCTURE EXPOSURE IMPACT

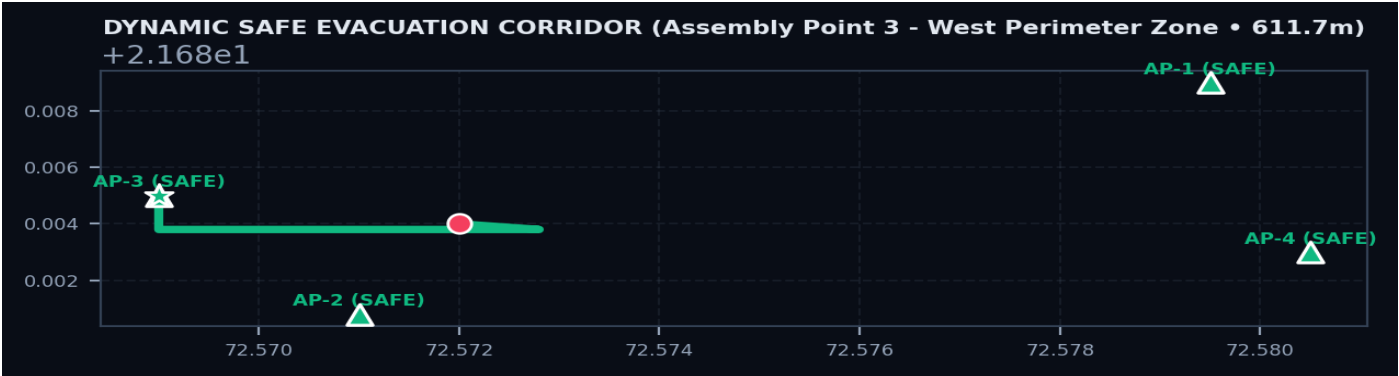
Total Workers on Site:	28 Personnel Exposure Assessment: No active seeded worker coordinates intersected the calculated threat envelopes at simulation time.	Lethal Zone (Red) Exposed:	0 Workers
Severe Zone (Orange) Exposed:	0 Workers	Caution Zone (Yellow) Exposed:	0 Workers
Compromised Plant Assets:	1 of 16 Units	Severed Road Segments:	2 of 10 Segments

4. DYNAMIC SAFE EVACUATION CORRIDOR & MUSTER DIRECTIVES

Designated Safe Assembly Point:	Assembly Point 3 - West Perimeter Zone	Exit Perimeter Gate:	Gate 2 - South Commercial & Tanker Logistics Gate
Total Safe Egress Distance:	611.7 meters	Estimated Walk Time:	8.5 minutes (at 1.2 m/s)
Multi-Factor Scoring:	Safety: 70% Dist: 61% Rank: 0.667	Route Clearance Status:	DIVERTED
Selection Rationale:	Selected as optimal Crosswind egress corridor (65° clearance from plume axis) providing shortest uncompromised route to Assembly Point 3 - West Perimeter Zone (Safety Score: 0.70, Composite Score: 0.67).	Avoided Blocked Roads:	Central Spine Avenue (North-South), Chemical Sector Cross Road (East-West North)

Evaluated Muster Point Alternatives & Rejection Analysis:

Candidate Assembly Point	Exit Gate	Distance	Safety	Status / Rejection Reason
ROUTE-AP-3 (AP-3)	GATE-2	611.7 m	70%	SELECTED: None
ROUTE-AP-4 (AP-4)	GATE-3	970.2 m	70%	VIABLE BACKUP: Internal access roads severed; requires outer peripheral detour
ROUTE-AP-1 (AP-1)	GATE-1	1356.1 m	70%	REJECTED: Internal access roads severed; requires outer peripheral detour
ROUTE-AP-2 (AP-2)	GATE-2	605.2 m	25%	REJECTED: Downwind of toxic plume (29° from plume axis); high vapor inhalation hazard



5. TACTICAL EMERGENCY RESOURCE DEPLOYMENT & SUPPRESSION STRATEGY

DECISION-SUPPORT RECOMMENDATION — REQUIRES SITE/HSE VALIDATION: All tactical quantities, PPE ensembles, suppression demands, and response actions are prototype computational recommendations and must be validated against facility ERDMP and competent safety authorities before operational deployment.

Upwind Staging Standoff:	250.0 meters	Isolation Cordon Radius:	398.0 meters
Firewater Demand:	7,300 LPM	Foam Concentrate Demand:	2,750 Liters (AFFF 3%)
Mandatory Entry PPE:	Flash-Fire Resistant Nomex III A Coveralls, Chemical Splash Shield, Level B SCBA	Calculation Basis:	Calculated from release rate 25.0 kg/s, 1 vulnerable assets, and Liquefied Petroleum Gas (LPG - Propane/Butane) SDS toxicity/flammability

Unit Name & Type	Station	Staging Geolocation	Transit	ETA	Priority
Hazmat Rapid Response Unit (Level A/B Ensembles) Hot Zone Isolation & Emergency Valve Clamping	FS-01	Incident Hot-Zone Control Post (225° Upwind)	110.1 m	0.8m	IMMEDIATE
Fire Tender Alpha (10,000L Multi-Purpose Water/Foam) Vapor Cloud Inerting & Preemptive Foam Blanket	FS-01	Upwind Incident Staging Post (Bearing 225°, 250m Standoff)	133.2 m	0.9m	IMMEDIATE
Industrial Emergency Response Team (ERT Alpha) Cordon Enforcement, Road Junction Barricading & Headcount Control	CR-01	Main Outer Access Gate & Security Command	303.6 m	1.3m	SUPPORT
High-Volume Water Curtain Bowser Secondary Water Supply & Runoff Dilution	FS-01	Crosswind Knockdown Sector (315° Vector)	350.2 m	1.5m	SUPPORT

Advanced Cardiac & Toxic Resuscitation Ambulance Precautionary Medical Standby at Muster Point	MED-01	Medical Triage at Assembly Point 3 - West Perimeter Zone	805.5 m	2.7m	STANDBY
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6. INCIDENT COMMAND EXECUTIVE SUMMARY SNAPSHOT

Risk:	44.6/100 — MODERATE	Chemical:	Liquefied Petroleum Gas (LPG - Propane/Butane)
Personnel Exposed:	0 workers (Red: 0)	Compromised Assets:	1 of 16 units
Blocked Roads:	2 segments	Primary Muster:	Assembly Point 3 - West Perimeter Zone
Exit Gate:	Gate 2 - South Commercial & Tanker Logistics Gate	Egress Metrics:	611.7 m / 8.5 min
Lead Tactical Resource:	Hazmat Rapid Response Unit (Level A/B Ensembles) (IMMEDIATE, ETA: 0.8m)	Firewater Demand:	7,300 LPM Water (2,750 L AFFF Foam)

7. STANDARD OPERATING PROCEDURE (SOP) PHASED ACTION DIRECTIVES

Phase 1: Initial Alarm, Isolation & Personnel Evacuation (0 - 5 Minutes) • Sound Plant Emergency Siren (Continuous Alarm for Liquefied Petroleum Gas (LPG - Propane/Butane) Tank Leak). • Actuate Emergency Shutdown (ESD) for T-03 and isolate upstream pipeline valves. • Enforce mandatory crosswind evacuation of all Sector personnel towards Assembly Point 3 - West Perimeter Zone. • ERT Team establishes 398.0m initial cordon perimeter around T-03. Phase 2: Source Mitigation & Tactical Exposure Protection (5 - 15 Minutes) • Position Fire Tender & Bowser at Upwind Staging Post (250m standoff, bearing 225°). • Establish 7,300 LPM water curtain coverage to suppress Liquefied Petroleum Gas (LPG - Propane/Butane) vapors and cool 1 adjacent assets. • Deploy Hazmat Entry Squad in Flash-Fire Resistant Nomex III A Coveralls, Chemical Splash Shield, Level B SCBA for hot-zone valve isolation and clamp seal. • Verify automated fixed water deluge systems on neighboring storage vessels. Phase 3: Containment, Medical Triage & Incident De-escalation (15 - 30 Minutes) • Complete headcount verification at Assembly Point 3 - West Perimeter Zone (Reconcile against 0 exposed workers). • Ambulance team initiates emergency triage and decontamination for exposed operators. • Contain all contaminated firefighting water in industrial effluent retention sump; prevent municipal drain escape. • HSE Lead Commander compiles situation briefing for District Disaster Management Authority (DDMA).
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8. DATA TRACEABILITY & SCREENING MODEL ASSUMPTIONS

Dispersion Physics:	Screening Gaussian Approximation with Pasquill-Gifford stability parameters	Human Governance:	Mandatory HSE human review and authorization required prior to operational execution
Weather Telemetry:	DEMO Telemetry (Scenario Preset) • Recorded at 34.0°C, 12.0 km/h	Statutory Scope:	Decision-support prototype only; non-certified under statutory OISD/ALOHA rules
Terrain Modeling:	Simplified 2D/3D flat terrain representation with standard surface roughness (z ₀ = 0.5m)	Road & Worker Graph:	Prototype facility CAD/GIS network with obstacle avoidance pathfinding

9. REGULATORY DISCLAIMER & PROTOTYPE NOTICE

PROTOTYPE DECISION SUPPORT — NON-CERTIFIED COMPUTATIONAL WORKFLOW:

This emergency pre-plan document is prepared automatically by the SIH 1505 Decision Support Engine. Dispersion envelopes, routing corridors, and tactical dispatches are computed for hackathon demonstration and decision-support prototyping only. This software does NOT claim certified ALOHA equivalence or statutory compliance. Operational tactical actions require validation against the facility's approved Emergency Response and Disaster Management Plan (ERDMP), applicable OISD/PESO standards, and competent statutory safety authorities.

10. HUMAN-IN-THE-LOOP PRE-PLAN AUTHORIZATION & ENDORSEMENT

DOCUMENT STATUS: AUTHORIZED (PROTOTYPE DEMO)

Document Version: v1.0

Prepared by: SIH-1505 Decision Support Engine

Demonstration Approver: Santosh Kadam

Role / Designation: Chief Fire Marshal

Authorization Record ID: AUTH-20260815-12057A

Authorized At: 2026-08-15 17:27:12 UTC

Approval Notes: LPG BLEVE Deluge Plan Authorized.

DEMONSTRATION SIGNATURE BLOCK:

✓ **AUTHORIZED (PROTOTYPE DEMO)**

Santosh Kadam

Chief Fire Marshal

DEMO SIGNATURE — NOT A REAL SIGNATURE

(Ready for PKI Digital Signature Integration)